

DANDI:000017: Distributed coding of choice, action and engagement across the mouse brain

Steinmetz, Nicholas; Zatzka-Haas, Peter; Carandini, Matteo; Harris, Kenneth; Wang, Renee (2024) Distributed coding of choice, action and engagement across the mouse brain (Version 0.240329.1926) [Data set]. DANDI archive. <https://doi.org/10.48324/dandi.000017/0.240329.1926>

Source labels: **DANDI metadata** = the Dandiset's published metadata; **NWB file** = values or text stored in the NWB files; **Computed by this export** = derived by the export scripts in `dandi_export/` (definitions given where used).

1. The dataset as published

Everything in this section is quoted from the Dandiset's published metadata or, for file-level fields, from `sub-Cori_i_ses-20161214T120000.nwb`.

Field	Value (DANDI metadata)
Name	Distributed coding of choice, action and engagement across the mouse brain
Identifier / version	DANDI:000017 / 0.240329.1926
Citation	Steinmetz, Nicholas; Zatzka-Haas, Peter; Carandini, Matteo; Harris, Kenneth; Wang, Renee (2024) Distributed coding of choice, action and engagement across the mouse brain (Version 0.240329.1926) [Data set]. DANDI archive. https://doi.org/10.48324/dandi.000017/0.240329.1926
DOI	10.48324/dandi.000017/0.240329.1926
Contributors (roles)	Steinmetz, Nicholas (Author, ContactPerson); Zatzka-Haas, Peter (Author); Carandini, Matteo (Author); Harris, Kenneth (Author); Wang, Renee (DataCurator)
License	CC-BY-4.0
Keywords	neuropixels
Related resources	DOI:10.1038/s41586-019-1787-x (IsDescribedBy)
Files / size / subjects	39 files, 14.68 GB, 10 subjects
Species	House mouse
Approach	electrophysiological approach; behavioral approach
Measurement technique	analytical technique; spike sorting technique; behavioral technique; surgical technique
Variables measured	ProcessingModule, PupilTracking, BehavioralEpochs, Units, BehavioralEvents, BehavioralTimeSeries, ElectrodeGroup
Date published	2024-03-29T19:26:55.755552+00:00

Description (DANDI metadata, verbatim)

Data from "Distributed coding of choice, action and engagement across the mouse brain" Steinmetz et. al Nature 2019

File-level fields (NWB file, verbatim)

Field	Value (NWB file)
session_description	Neuropixels recording during visual discrimination in awake mice.
experiment_description	Large-scale Neuropixels recordings across brain regions of mice during a head-fixed visual discrimination task.
institution	University College London
lab	The Carandini and Harris Lab
experimenter	Nick Steinmetz
keywords	Neural coding; Neuropixels; mouse; brain-wide; vision; visual discrimination; electrophysiology
related_publications	DOI 10.1038/s41586-019-1787-x
subject	subject_id: Cori; species: Mus musculus; genotype: tetO-G6s x CaMK-tTA; sex: F; age: 77 days; description: strain: C57Bl6/J

Field	Value (NWB file)
electrode_groups	Probe1: description = 'Neuropixels Phase3A opt3'; Probe2: description = 'Neuropixels Phase3A opt3'

2. Contents of the files

Quantity	Value	Source
Sessions / subjects	39 / 10	NWB file
Subjects (sessions each)	Cori (3), Forssmann (4), Hench (4), Lederberg (7), Moniz (3), Muller (3), Radnitz (5), Richards (5), Tatum (4), Theiler (1)	NWB file
Species	Mus musculus (subject.species)	NWB file
Strain (subject.description)	strain: C57Bl6/J	NWB file
Genotypes (subject.genotype)	Ai95 x VGlut1-Cre; C57BL/6J; Snap25-GCaMP6s; Vglut1-Cre; tetO-G6s x CaMK-tTA	NWB file
Sex, by subject	F: 6, M: 4	NWB file
Age at session (subject.age)	77–322 days	NWB file
Electrode groups	Probe1, Probe2, Probe3; description 'Neuropixels Phase3A opt3'	NWB file
Unit sampling_rate	30000 Hz	NWB file
Trials (included == True)	10,050 (9,538)	NWB file
Visual contrast values	0, 0.25, 0.5, 1	NWB file
response_choice values	-1, 0, 1	NWB file
feedback_type values	-1, 1	NWB file
Units (phy_annotation ≥ 2)	48,661 (33,997)	NWB file
phy_annotation counts	1: 14,664, 2: 10,314, 3: 23,683	NWB file
Distinct electrode 'location' labels at unit peak channels	73 (including 'root')	NWB file
Total spikes	442,033,914	NWB file
'spontaneous' interval time per session	median 882 s (range 234–1261 s)	Computed by this export
Recording duration	35.1 h total	Computed by this export

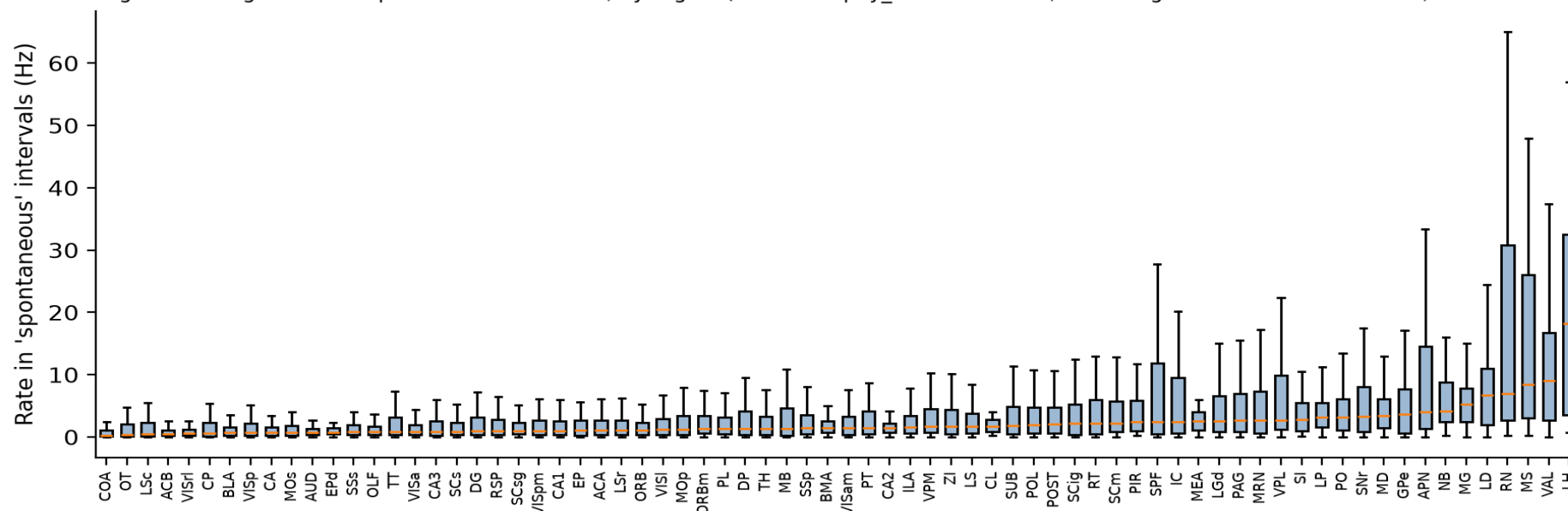
Definitions of phy_annotation, the trial columns, the 'spontaneous' intervals and the electrode columns are quoted in section 8.

3. Export parameters

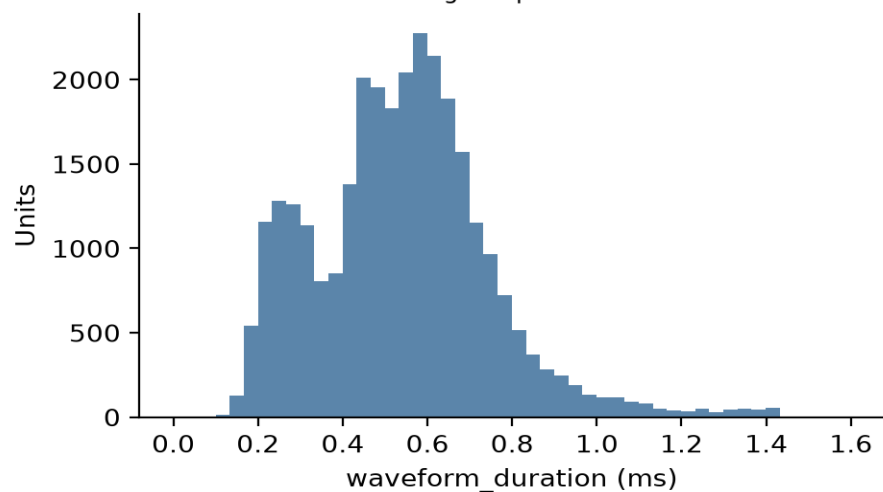
Parameter	Definition	Source
Unit region	electrodes.location at row peak_channel – 1 (peak_channel is 1-based; the first electrode listed for each unit is always peak_channel – 1)	Computed by this export
good	phy_annotation ≥ 2 (the inclusion rule stated in the phy_annotations description)	NWB file
mean_rate_hz	n_spikes / recording duration	Computed by this export
task_rate_hz	spikes between first trial start and last trial stop / that duration	Computed by this export
spontaneous_rate_hz	spikes inside the intervals/spontaneous table / total duration of those intervals	Computed by this export
waveform_duration_ms	waveform_duration (samples) / sampling_rate × 1000; see section 7 for the 11 sessions where it was recomputed	Computed by this export
count_prestim	spikes from -0.5 to +0 s relative to trials.visual_stimulus_time	Computed by this export
count_stim	spikes from +0 to +0.25 s relative to trials.visual_stimulus_time	Computed by this export
count_remove	spikes from -0.25 to +0 s relative to trials.response_time	Computed by this export
count_feedback	spikes from +0 to +0.5 s relative to trials.feedback_time	Computed by this export
wheel_position export	downsampled from 2500 Hz to 100 Hz (every 25th sample)	Computed by this export

4. Overview figures (Computed by this export)

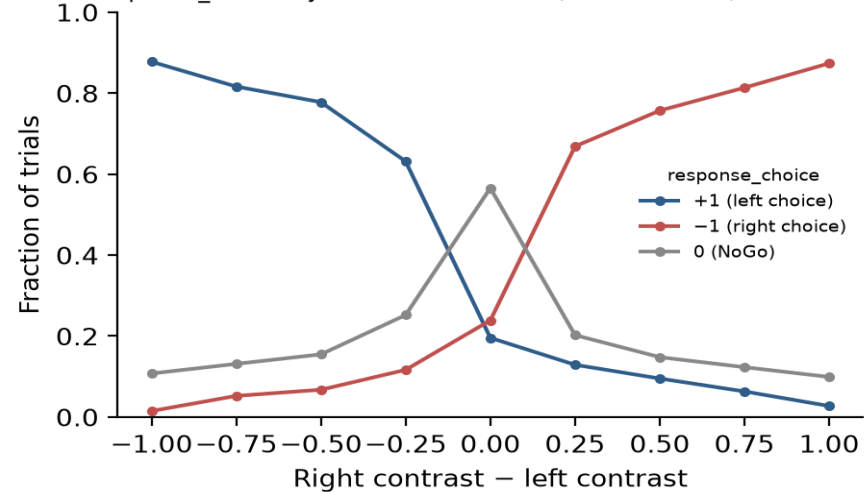
Firing rate during the NWB 'spontaneous' intervals, by region (units with phy_annotation ≥ 2 ; all 72 regions with ≥ 20 such units; outliers hidden)



Distribution of trough-to-peak waveform duration



response_choice by contrast difference (included trials, all sessions)



5. All regions (Computed by this export)

One row per electrode 'location' label, for units with phy_annotation ≥ 2 , excluding 'root'. Sorted by number of units. Region labels are exactly as stored in the NWB electrodes table.

Region (location)	Units	Sessions	Spontaneous-interval rate mean (Hz)	SD (Hz)	Median (Hz)	Whole-recording rate mean (Hz)	waveform_duration median (ms)
MOs	1743	18	1.51	2.36	0.74	1.43	0.63
CA1	1360	21	2.19	3.39	1.03	2.18	0.63
TH	1257	8	2.77	4.10	1.36	2.75	0.43
CP	1215	5	2.42	4.53	0.59	2.51	0.63
VIsp	1118	12	1.88	3.04	0.70	2.03	0.60
MRN	1019	11	7.06	11.65	2.73	7.41	0.30
VISam	1001	11	2.86	4.02	1.46	2.88	0.60
LGd	882	10	4.79	6.59	2.62	4.44	0.47
MB	874	6	5.45	11.45	1.38	5.07	0.33
PL	863	10	2.64	3.73	1.31	2.42	0.63
ACA	841	11	2.09	2.83	1.07	1.88	0.63
ORB	813	5	1.86	2.50	1.12	1.67	0.60
SCig	812	4	4.38	7.43	2.19	4.53	0.27
MOp	793	3	3.26	5.25	1.28	3.56	0.57
DG	776	16	3.08	6.64	0.95	3.26	0.50
RSP	770	9	2.03	2.97	0.96	1.86	0.57
LP	748	11	4.26	4.36	3.16	4.04	0.50
PO	692	5	4.18	4.10	3.19	4.47	0.47
SUB	685	9	3.90	5.41	1.85	3.91	0.57
VISpm	589	4	2.10	3.17	1.03	2.14	0.60
SSp	515	5	3.33	5.36	1.42	3.53	0.57
CA3	494	10	2.47	4.52	0.88	2.63	0.60
LSr	475	4	2.06	2.66	1.09	2.07	0.53
VISI	434	3	2.55	3.82	1.19	2.75	0.57
ILA	411	3	2.83	3.52	1.57	2.52	0.63
LS	405	3	3.50	5.87	1.71	3.60	0.50
MD	397	3	4.61	4.51	3.47	4.89	0.53
SCm	356	7	5.13	8.11	2.24	5.06	0.27
VPM	353	4	3.25	3.92	1.66	3.35	0.43
VISa	340	5	1.82	2.64	0.86	1.82	0.60
ACB	337	3	1.29	2.24	0.50	1.51	0.63
BLA	317	2	1.47	2.85	0.70	1.51	0.63
COA	316	1	0.83	1.43	0.22	0.96	0.60
POST	312	4	3.51	4.36	2.07	3.45	0.50
SNr	308	4	8.20	14.07	3.25	8.45	0.27
VPL	301	4	6.52	8.65	2.74	6.92	0.43
LD	298	6	7.63	6.86	6.72	7.73	0.47
MG	295	2	5.80	4.44	5.27	5.88	0.43
GPe	292	3	7.83	15.67	3.70	7.92	0.30

Region (location)	Units	Sessions	Spontaneous-interval rate mean (Hz)	SD (Hz)	Median (Hz)	Whole-recording rate mean (Hz)	waveform_duration median (ms)
ZI	289	4	4.19	7.21	1.69	4.72	0.43
VISrl	286	2	0.91	1.19	0.55	0.76	0.63
OLF	263	3	1.56	2.98	0.80	1.41	0.60
APN	247	2	11.11	15.06	4.01	11.28	0.27
AUD	246	1	1.11	1.24	0.77	1.16	0.60
POL	215	3	4.36	7.59	2.00	4.54	0.43
CA	193	1	1.67	3.96	0.73	1.70	0.67
TT	181	4	2.86	6.12	0.83	2.75	0.57
SCsg	178	6	2.26	3.42	0.96	2.42	0.40
RT	173	2	3.96	4.51	2.20	4.08	0.27
SCs	157	4	2.14	3.31	0.90	2.69	0.27
PAG	141	3	5.18	6.63	2.63	4.74	0.23
ORBm	122	1	2.69	3.78	1.30	2.68	0.67
SSs	120	1	1.42	1.63	0.80	1.61	0.53
CA2	115	1	1.93	2.20	1.52	1.89	0.40
BMA	104	1	1.93	1.66	1.45	1.87	0.57
EP	91	1	1.77	2.08	1.05	1.54	0.43
VAL	83	1	11.75	11.21	9.07	11.85	0.43
LSc	72	1	2.06	3.30	0.45	1.96	0.50
PIR	67	1	4.54	5.23	2.39	4.37	0.57
IC	58	1	7.57	11.27	2.50	8.19	0.22
RN	58	1	16.72	20.62	6.92	18.11	0.27
SPF	53	2	9.67	15.15	2.40	10.03	0.37
EPd	52	1	1.23	1.77	0.78	1.04	0.63
SI	48	1	8.85	16.66	2.79	8.51	0.30
PT	45	1	3.45	4.61	1.46	3.75	0.50
NB	43	1	6.81	8.42	4.19	6.41	0.27
MEA	41	1	3.62	5.48	2.60	3.30	0.53
DP	39	1	3.83	6.24	1.33	3.20	0.60
CL	32	1	2.52	3.24	1.76	2.12	0.57
OT	26	1	1.27	1.65	0.41	1.02	0.62
LH	24	2	20.31	17.93	18.16	16.61	0.40
MS	23	1	17.26	19.24	8.40	17.28	0.33

6. Sessions

Session	Sex	Age	Genotype	Duration (s)	Trials	Units	phy ≥ 2	feedback_type = 1 (%)	NoGo (%)	Median response time – go_cue (s)	Regions
Cori_2016-12-14	F	77 days	tetO-G6s x CaMK-tTA	2703	214	1085	734	65.9	34.6	0.521	ACA, CA3, DG, LS, MOs, SUB, VISp, root
Cori_2016-12-17	F	84 days	tetO-G6s x CaMK-tTA	3147	251	1146	1070	63.3	37.5	0.509	CA1, POST, VISI, VISpm, root
Cori_2016-12-18	F	84 days	tetO-G6s x CaMK-tTA	3834	228	1089	619	66.2	33.8	0.441	CA1, DG, LP, MG, MRN, NB, POST, SPF, VISam, VISp, root
Forssmann_2017-11-01	M	182 days	C57BL/6J	3115	249	2073	1769	66.7	50.6	1.797	ACA, CA1, DG, LGd, LSr, MOs, SUB, TH, VISa, VISp, VPL
Forssmann_2017-11-02	M	182 days	C57BL/6J	3044	254	1569	1077	66.1	40.9	0.470	ACA, CA1, DG, MOs, OLF, ORB, PL, SUB, VISa, root
Forssmann_2017-11-04	M	189 days	C57BL/6J	3472	290	1435	1169	74.1	31.7	0.378	AUD, CA1, SSp, TH, root
Forssmann_2017-11-05	M	189 days	C57BL/6J	3123	252	1029	584	67.1	24.6	0.385	CA3, CP, EPd, LD, PIR, SSp, VPL, root
Hench_2017-06-15	M	217 days	tetO-G6s x CaMK-tTA	3260	250	1707	1157	64.4	37.2	0.461	CA1, CA3, DG, ILA, LD, LP, LSr, MOs, PL, PO, SUB, TT, VISa, VISp, root
Hench_2017-06-16	M	224 days	tetO-G6s x CaMK-tTA	3915	372	1364	788	68.5	36.0	0.309	CA1, CA3, LD, LSr, ORBm, PL, TH, TT, VISam, VISI, VPL, root
Hench_2017-06-17	M	224 days	tetO-G6s x CaMK-tTA	4083	447	1591	1172	62.0	34.7	0.355	CA1, DG, GPe, MB, MRN, POL, POST, SCm, SCsg, VISI, VISp, VISrl, root
Hench_2017-06-18	M	224 days	tetO-G6s x CaMK-tTA	3720	342	1409	857	79.5	28.4	0.337	CP, LSc, LSr, MOp, PT, root
Lederberg_2017-12-05	F	105 days	Vglut1-Cre	3446	340	1219	698	73.8	18.8	0.179	ACA, CA1, DG, LGd, LH, MD, MOs, PL, SUB, VISam, VISp, root
Lederberg_2017-12-06	F	105 days	Vglut1-Cre	3387	300	1262	983	79.7	18.7	0.192	ACA, CA1, DG, LGd, MB, MOs, MRN, MS, PL, RN, SCm, SCs, VISam, ZI, root
Lederberg_2017-12-07	F	105 days	Vglut1-Cre	3195	268	1228	756	69.4	22.8	0.270	CA1, MOs, MRN, ORB, PAG, RSP, SCm, SCs, VISp, root
Lederberg_2017-12-08	F	105 days	Vglut1-Cre	3582	404	1134	743	76.5	24.0	0.265	BLA, CA3, GPe, LGd, MB, VPM, ZI, root
Lederberg_2017-12-09	F	105 days	Vglut1-Cre	3128	280	977	474	71.8	26.4	0.339	CA3, LGd, MB, SSp, SSs, TH, root
Lederberg_2017-12-10	F	105 days	Vglut1-Cre	2894	224	727	565	83.0	23.2	0.260	LD, MEA, RT, VPL, VPM, root
Lederberg_2017-12-11	F	105 days	Vglut1-Cre	3497	316	1350	1090	78.2	24.1	0.186	ACB, CA3, CP, LGd, OT, SI, SNr, TH, ZI, root
Moniz_2017-05-15	M	210 days	Ai95 x VGlut1-Cre	3229	247	806	606	57.5	45.3	1.139	APN, CA1, DG, LP, POL, SUB, VISam, VISpm, root
Moniz_2017-05-16	M	210 days	Ai95 x VGlut1-Cre	3037	235	1144	899	59.1	40.0	0.635	CA1, DG, LGd, LP, MG, SPF, SUB, TH, VISa, VISp, root
Moniz_2017-05-18	M	210 days	Ai95 x VGlut1-Cre	2761	124	795	578	53.2	45.2	0.730	LD, MOp, SSp, VAL, root
Muller_2017-01-07	F	119 days	Ai95 x VGlut1-Cre	4078	444	1109	646	70.0	25.9	0.340	ACA, ACB, IC, MOs, MRN, PL, RSP, SCsg, TT, VISp, root
Muller_2017-01-08	F	119 days	Ai95 x VGlut1-Cre	2820	151	1455	1270	67.5	43.0	0.768	CA1, DG, LP, PO, VISam, root
Muller_2017-01-09	F	119 days	Ai95 x VGlut1-Cre	2983	187	1426	1364	58.3	44.4	0.696	CA1, CA2, MRN, RSP, SCig, SCsg, VISpm, VISrl, root

Session	Sex	Age	Genotype	Duration (s)	Trials	Units	phy ≥ 2	feedback_type = 1 (%)	NoGo (%)	Median response_time - go_cue (s)	Regions
Radnitz_2017-01-08	F	140 days	Snap25-GCaMP6s	3319	261	1266	885	72.0	29.1	0.240	ACA, DP, ILA, MOs, MRN, PL, RSP, SCig, SCsg, TT, VISp, root
Radnitz_2017-01-09	F	140 days	Snap25-GCaMP6s	3028	178	1161	1056	73.0	28.1	0.240	MOs, MRN, SCig, VISp, root
Radnitz_2017-01-10	F	140 days	Snap25-GCaMP6s	3657	253	1251	565	62.8	32.0	0.415	ACA, ACB, CA1, DG, LD, LP, MOs, PL, PO, VISam, root
Radnitz_2017-01-11	F	140 days	Snap25-GCaMP6s	2847	142	860	549	59.9	44.4	0.851	APN, CA1, DG, LP, MRN, POL, RSP, SCig, SCsg, VISpm, root
Radnitz_2017-01-12	F	140 days	Snap25-GCaMP6s	3156	128	1147	750	58.6	50.0	1.500	CP, MOp, SSp, root
Richards_2017-10-29	M	182 days	C57BL/6J	2648	143	1461	942	50.3	55.9	1.500	ACA, CA1, CL, DG, LP, LS, MD, MOs, PO, RSP, SUB, VISa, VPM, root
Richards_2017-10-30	M	182 days	C57BL/6J	2999	237	1251	979	69.2	32.1	0.339	CA3, MOs, OLF, ORB, POST, SCm, SNr, TH
Richards_2017-10-31	M	182 days	C57BL/6J	2968	260	778	522	71.5	33.1	0.327	MOs, MRN, OLF, ORB, PAG, RSP, SCm, SCs, root
Richards_2017-11-01	M	182 days	C57BL/6J	2761	191	854	613	67.5	29.8	0.290	CA1, CA3, CP, EP, LGd, SNr, VPM, ZI, root
Richards_2017-11-02	M	182 days	C57BL/6J	3242	296	928	781	59.5	47.6	1.125	BMA, COA, LP, MB, PO, RT, root
Tatum_2017-12-06	F	105 days	Vglut1-Cre	3432	311	1425	795	77.2	25.4	0.245	ACA, CA1, DG, LGd, LH, LP, LS, MD, MOs, RSP, TH, VISam, root
Tatum_2017-12-07	F	105 days	Vglut1-Cre	3203	258	1409	682	72.9	35.3	0.590	CA, MOs, MRN, PL, SCm, SCsg, VISam, root
Tatum_2017-12-08	F	105 days	Vglut1-Cre	3339	181	1907	1057	72.9	35.9	0.486	MB, MOs, ORB, PAG, RSP, SCm, SCs, VISam, root
Tatum_2017-12-09	F	105 days	Vglut1-Cre	2987	199	1041	805	65.8	36.7	0.536	BLA, CA3, GPe, LGd, SNr, root
Theiler_2017-10-11	F	322 days	Ai95 x VGlut1-Cre	3222	343	1793	1348	71.7	32.1	0.446	ACA, CA1, DG, ILA, LP, MOs, PL, SUB, VISam, VISp, root

Subject, genotype, age and counts: NWB file. Percentages and response times: computed by this export.

7. Data checks performed by this export

waveform_duration. In 12 of 39 sessions the NWB column *waveform_duration* correlates with *cluster_depths* at $r > 0.9999$ and reaches values up to ~3800 samples (~127 ms), i.e. it contains the depth values. For those sessions the export recomputes trough-to-peak duration from *waveform_mean* on the highest-amplitude channel. In sessions where the stored value is plausible, this recomputation agrees with it at $r \approx 0.96$. *units.csv* records which source was used (*waveform_duration_source*).

Sampling-rate field. The *wheel_position* series stores 0.0004 in its *rate* field, which is the sampling period (2500 Hz). The export treats rate values < 1 as periods.

Pupil timestamps. In some sessions the eye tracking series stores only two timestamps (first and last frame). The export spaces the frames evenly between them.

8. Column and signal descriptions from the NWB files

Quoted verbatim from sub-Cor1_ses-20161214T120000.nwb; all files in the Dandiset share this structure.

Location in NWB file	Name	Description (as written in the NWB file)
electrodes	(table)	metadata about extracellular electrodes
electrodes	x	the x coordinate of the channel location
electrodes	y	the y coordinate of the channel location
electrodes	z	the z coordinate of the channel location
electrodes	imp	the impedance of the channel
electrodes	location	the location of channel within the subject e.g. brain region
electrodes	filtering	description of hardware filtering
electrodes	group	a reference to the ElectrodeGroup this electrode is a part of
electrodes	group_name	the name of the ElectrodeGroup this electrode is a part of
electrodes	site_id	The site number, in within-probe numbering, of the channel (in practice for this dataset this always starts at zero and counts up to 383 on each probe so is equivalent to the channel number - but if switches had been used, the site number could have been different than the channel number).
electrodes	site_position	The x- and y-position of the site relative to the face of the probe (where the first column is across the face of the probe laterally and the second is the position along the length of the probe; the sites nearest the tip have second column=0).
electrodes	ccf_ap	The AP position in Allen Institute's Common Coordinate Framework.
electrodes	ccf_dv	The DV position in Allen Institute's Common Coordinate Framework.
electrodes	ccf_lr	The LR position in Allen Institute's Common Coordinate Framework.
electrodes	entry_point_rl	mediolateral position of probe entry point relative to midline (microns). Positive means right
electrodes	entry_point_ap	anteroposterior position of probe entry point relative to bregma (microns). Positive means anterior
electrodes	vertical_angle	vertical angle of probe (degrees). Zero means horizontal. Positive means pointing down
electrodes	horizontal_angle	horizontal angle of probe (degrees), after vertical rotation. Zero means anterior. Positive means counterclockwise (i.e. left).
electrodes	axial_angle	axial angle of probe (degrees). Zero means that without vertical and horizontal rotations, the probe contacts would be pointing up. Positive means "counterclockwise.
electrodes	distance_advanced	How far the probe was moved forward from its entry point. (microns).
units	(table)	Autogenerated by NWBFile
units	peak_channel	The channel number of the location of the peak of the cluster's waveform.
units	waveform_duration	The trough-to-peak duration of the waveform on the peak channel.
units	phy_annotations	0 = noise (these are already excluded and don't appear in this dataset at all); 1 = MUA (i.e. presumed to contain spikes from multiple neurons; these are not analyzed in any analyses in the paper); 2 = Good (manually labeled); 3 = Unsorted. In this dataset 'Good' was applied in a few but not all datasets to included neurons, so in general the neurons with _phy_annotation>=2 are the ones that should be included.
units	cluster_depths	The position of the center of mass of the template of the cluster, relative to the probe. The deepest channel on the probe is depth=0, and the most superficial is depth=3820. Units: μm
units	sampling_rate	Sampling rate, in Hz.
units	spike_times	the spike times for each unit
units	electrodes	the electrodes that each spike unit came from
units	electrode_group	the electrode group that each spike unit came from
units	waveform_mean	the spike waveform mean for each spike unit
units	spike_amps	The peak-to-trough amplitude, obtained from the template and template-scaling amplitude returned by Kilosort (not from the raw data).
units	spike_depths	The position of the center of mass of the spike on the probe, determined from the principal component features returned by Kilosort. The deepest channel on the probe is depth=0, and the most superficial is depth=3820.
intervals/spontaneous	(table)	Intervals of sufficient duration when nothing else is going on (no task or stimulus presentation)
intervals/spontaneous	start_time	Start time of epoch, in seconds

Location in NWB file	Name	Description (as written in the NWB file)
intervals/spontaneous	stop_time	Stop time of epoch, in seconds
intervals/trials	(table)	experimental trials
intervals/trials	start_time	Start time of epoch, in seconds
intervals/trials	stop_time	Stop time of epoch, in seconds
intervals/trials	included	Importantly, while this variable gives inclusion criteria according to the definition of disengagement (see manuscript Methods), it does not give inclusion criteria based on the time of response, as used for most analyses in the paper.
intervals/trials	go_cue	The 'goCue' is referred to as the 'auditory tone cue' in the manuscript.
intervals/trials	visual_stimulus_time	Times are relative to the same time base as every other time in the dataset, not to the start of the trial.
intervals/trials	visual_stimulus_left_contrast	Proportion contrast. A value of 0.5 means 50% contrast. 0 is a blank screen: no change to any pixel values on that side (completely undetectable).
intervals/trials	visual_stimulus_right_contrast	Proportion contrast. A value of 0.5 means 50% contrast. 0 is a blank screen: no change to any pixel values on that side (completely undetectable).
intervals/trials	response_time	Times are relative to the same time base as every other time in the dataset, not to the start of the trial.
intervals/trials	response_choice	Enumerated type. The response registered at the end of the trial, which determines the feedback according to the contrast condition. Note that in a small percentage of cases (~4%, see manuscript Methods) the initial wheel turn was in the opposite direction. -1 for Right choice (i.e. correct when stimuli are on the right); +1 for left choice; 0 for Nogo choice.
intervals/trials	feedback_time	Times are relative to the same time base as every other time in the dataset, not to the start of the trial.
intervals/trials	feedback_type	Enumerated type. -1 for negative feedback (white noise burst); +1 for positive feedback (water reward delivery).
intervals/trials	rep_num	Trials are repeated if they are "easy" trials (high contrast stimuli with large difference between the two sides, or the blank screen condition) and this keeps track of how many times the current trial's condition has been repeated.
acquisition	lickPiezo	Voltage values from a thin-film piezo connected to the lick spout, so that values are proportional to deflection of the spout and licks can be detected as peaks of the signal.
acquisition	wheel_position	The position reading of the rotary encoder attached to the rubber wheel that the mouse pushes left and right with his forelimbs.
processing/behavior	(module)	behavior module
processing/behavior/BehavioralEpochs	wheel_moves	Detected wheel movements.
processing/behavior/BehavioralEvents	lick_times	Extracted times of licks, from the lickPiezo signal.
processing/behavior/BehavioralTimeSeries	face_motion_energy	Features extracted from the video of the frontal aspect of the subject, including the subject's face and forearms.
processing/behavior/PupilTracking	eye_area	Features extracted from the video of the right eye.
processing/behavior/PupilTracking	eye_xy_positions	Features extracted from the video of the right eye.

9. Exported files

File	Rows	Columns
tables/sessions.csv	39	session_id, file, subject_id, sex, age, genotype, strain, session_start_time, duration_s, n_trials, n_units, n_good_units, regions, spontaneous_duration_s
tables/units.csv	48,661	session_id, unit_id, region, probe, ccf_ap, ccf_dv, ccf_lr, peak_channel, depth_um, phy_annotation, good, waveform_duration_ms, waveform_duration_source, n_spikes, mean_rate_hz, task_rate_hz, spontaneous_rate_hz
tables/trials.csv	10,050	session_id, trial_id, start_time, stop_time, included, go_cue, visual_stimulus_time, visual_stimulus_left_contrast, visual_stimulus_right_contrast, response_time, response_choice, feedback_time, feedback_type, rep_num
tables/region_summary.csv	72	region, n_units, n_sessions, spont_rate_mean_hz, spont_rate_sd_hz, spont_rate_median_hz, mean_rate_mean_hz, waveform_duration_ms_median
tables/dataset_metadata.json, nwb_file_metadata.json, nwb_descriptions.csv		the published metadata quoted in sections 1 and 8
trial_unit_counts.parquet	trials × units per session	session_id, trial_id, unit_id, count_prestim, count_stim, count_remove, count_feedback
spikes/<session>.parquet	442,033,914 total	unit_id, time (s), amp, depth
behavior/<session>__<signal>.parquet	per signal	wheel_position, wheel_moves (start_time, stop_time, type), lick_times, eye_area, eye_xy_positions, face_motion_energy